Worksheet 5-3 A Intervals of Increase and Decrease #1
AP Calculus Name_____

Directions: Given the following, find the critical values of the function $f(x)$. If $f(x)$ does not have any critical values, write "$f(x)$ has no critical values".

1. $f'(x) = 4\cos(x) - 2$ on $[0, 2\pi)$

2. $f(x) = \frac{e^{2x}}{x-3}$

3. $f(x) = x^2 \ln(x)$

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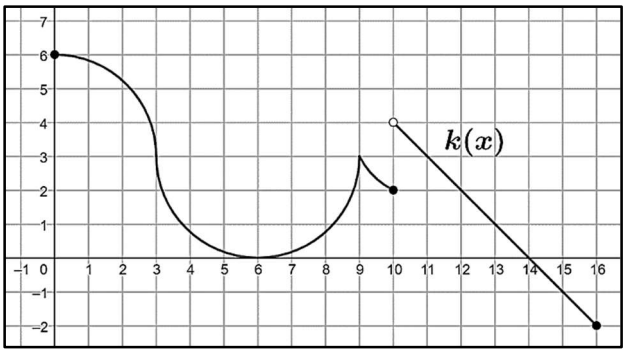
4. The graph of $k(x)$ is given at the right. Find any x values in the open interval $(0, 16)$ where $k(x)$ has a critical value.



Directions: Given the following, find the intervals where the function $g(x)$ is increasing or decreasing.

5. $g(x) = \sqrt[3]{x^2 - 4}$

6. $g'(x) = (x-3)e^{x+2}$

7. $g(x) = \frac{x+3}{x-4}$

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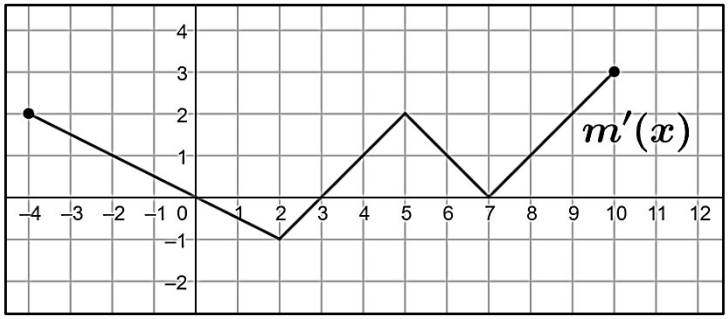
8. The graph of $m'(x)$ is given at the right. On what intervals is the graph of $m'(x)$ increasing? Explain your reasoning.



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9. Let g be a differentiable function such that $g'(x) < 0$ for all values of x and let $h(x) = g(x^2 - 2x - 8)$. Find the intervals where $h(x)$ is increasing and where $h(x)$ is decreasing. Explain your reasoning.